Chapter 6

The Perceptual to Conceptual Shift in Infancy and Early Childhood: A Surface or Deep Distinction?

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Categorization abilities in infancy are perceptual skills, based on the correlations between object features. Categorization skills in childhood also involve perceptual analyses of objects, but in addition are based on naïve theories about category membership and knowledge about characteristic or defining category features. The key point is whether we should interpret this to mean that categorization in infancy is qualitatively different from categorization in childhood.

—Goswami (1998, p. 289)

INTRODUCTION

Of all the ongoing debates within the early cognitive developmental literature, perhaps none is more hotly disputed and ill-defined than the transition known as the *perceptual to conceptual shift.* This phrase is generally used to describe the phenomenon whereby younger children and infants tend to rely on perceptual properties as the basis for categorization, whereas older children, typically toddlers and preschoolers, weigh more heavily the nonobvious or conceptual properties of objects in category membership decisions (Gelman & Markman, 1986, 1987; Mandler, 1992; Keil, 1991; Murphy & Medin, 1985; Piaget, 1952; Wellman & Gelman, 1988). In other words, it is believed that to determine which things "belong together," infants use surface properties such as facial features, shape, parts, or texture, but older children and presumably adults use deeper properties that are

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less often available in the perceptual array such as labels, internal biological components, or motion capabilities.

A classic example of this developmental phenomenon can be found in a study by Keil (1989; see also Rips, 1989) who presented children with stories in which various objects underwent an appearance transformation but retained their nonobvious properties. In one such story, children were shown a picture of a raccoon and told that it possessed the blood, bones, and reproductive properties of a skunk. From around 4 years of age, but not earlier, children judged that in such cases it was the nonobvious properties rather than the perceptual appearance of things that determine what they are, or more specifically, how they should be labeled. A similarly powerful illustration comes from work by Gelman and colleagues (e.g., Gelman & Markman, 1986, 1987; Davidson & Gelman, 1990) who found that 2½- to 3-year-old children, but not those around 2 years of age, make inductions to perceptually dissimilar pictures of objects of the same category (e.g., a cobra after being shown a brown snake) rather than to perceptually similar objects from a different category (e.g., a worm). Other examples that attest to young children's appreciation for nonobvious properties abound in the cognitive developmental literature. It has been found, for instance, that by preschool age children become sensitive to invisible properties such as germs (Au, Sidle, & Rollins, 1993; Kalish, 1996), aspects of the reproductive process (Springer, 1996), and features of growth, inheritance, and illness (e.g., Backscheider, Shatz, & Gelman, 1993; Springer & Keil, 1991). These data are often used to support the *theory view*, by which is meant that children of preschool age and older categorize and make inferences on the basis of information about ontology, causation, function, and other properties that are not directly observable (Carey, 1985; Gelman & Koenig, 2003; Gopnik & Wellman, 1994).

In light of recent work with infants it might also be argued that a similar transition occurs prior to the preschool years with regard to the knowledge acquisition of motion properties. For example, infants as young as 3 months can form categorical representations for cats, dogs, people, animals, and furniture and it is assumed that this behavior is grounded in perceptual information alone (for reviews see Quinn, chap. 5, this volume; Quinn & Eimas, 1996a, 1997). However, according to some researchers, by around 9 to 11 months of age infants categorize as equivalent those same classes of objects through *conceptual* categorization whereby things are grouped because they share a *meaning* (e.g., grouping animals because they are self-propelled) (Mandler & Bauer, 1988; Mandler, Bauer, & McDonough, 1991; Mandler & McDonough, 1998a). Moreover, it has been suggested that infants in the same age range can make inductive inferences such that they can generalize behaviors like drinking and sleeping to animals rather than to vehicles, and behaviors like starting with a key and giving people rides to

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vehicles rather than to animals (Mandler & McDonough, 1996, 1998b). Is it possible that this development in infancy reflects the same kind of shift observed in preschoolers?

The developmental change just outlined—namely, the acquisition of, and reliance on, information not consistently available in the perceptual array—is clearly a landmark in infants' and young children's evolving representations for objects in the world. Over and above being able to categorize things on the basis of their surface appearance, children following this transition can appropriately attribute various biological, psychological, and physical characteristics to different object kinds. But why, one might ask, is this latter state preferable? Perceptual cues, it is argued, are inherently unreliable as the basis for categorization because they are generally epiphenomenal in relation to the "true" or essential properties that characterize category membership. Perceptually given features such as "has four legs" or "has a long tail" are thought to be misleading in category membership decisions—not all dogs have a long tail—but deeper, nonobvious properties tend not to be (e.g., all dogs have hearts and dog DNA) (Murphy & Medin, 1985). Moreover, it is assumed that representations that include nonobvious information of objects are more abstract and more complex, which ultimately allows them more flexibly to be applied in cognitive functions such as induction, language, and memory.

Despite this portrayal of the tendency toward conceptual categorization in early development, things are not quite so cut and dry. I refer not to minor disputes over the interpretation of children's performance in one task or another but rather a questioning of the fundamental definitions of the terms involved and the content and structure of early representations. Relatively recent advances in developmental experimental methodology as well as the emergence of new theoretical perspectives, for example, have allowed researchers to examine more directly the nature of perceptual and conceptual categorization (e.g., Eimas, 1994; Jones & Smith, 1993; Oakes & Madole, 1999; Quinn & Eimas, 1997; Rakison & Poulin-Dubois, 2001; Rakison, 2003). At issue is not only the nature of conceptual structure but also its content; that is, the form in which concepts are represented, and the information about things in the world that is included within these concepts. The questions posed are, therefore, not trivial and are put forward mainly at a theoretical level. As will become apparent, there are also currently wildly diverging views on each of these issues within the developmental literature, and there are data that can be taken to support each side of the argument.

For the purposes of confining the scope of the chapter I focus predominantly on five separate but very much interrelated issues that pertain to the *perceptual to conceptual shift* (hereafter labeled the P2CS). The themes to be addressed are:

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1. What is the difference between perceptual and conceptual information;
2. Whether children undergo a stage-like "perceptual to conceptual" shift;
3. Whether children rely on perceptual information rather than conceptual information in category membership decisions;
4. Whether perceptual information is inherently less reliable than nonobvious properties as a basis for categorization; and
5. What mechanism or mechanisms underlie the acquisition of knowledge about perceptual properties and nonobvious properties?

By concentrating on these themes my aim is to provide a detailed description of the P2CS, to discuss the pivotal claims concerning the nature of representational change in infancy and early childhood, and to supply a (hopefully) cohesive theoretical perspective on what is considered to be a fundamental yet rather opaque area of cognitive development.

What Is Perceptual Information and Conceptual Information?

One of the problems with the notion of the perceptual to conceptual shift is that the very terms *perceptual* and *conceptual* are tricky to delineate, and in many cases individual researchers characterize them quite differently. Generally, the terms "perceptual information" and "perceptual cues" refer to properties or features of an object, entity, or event that are consistently available in the sensory array. Thus perceptual information is often used to describe the surface properties or features of things such as size, color, shape, parts, and so on. In contrast, the terms "conceptual," "nonobvious," or "nonobservable" information generally refer to those properties or features of an object, entity, or event that are not often if at all available in the sensory array. Such properties are generally taken to include internal biological and technological components of animates and inanimates (e.g., brains, hearts, engines, silicon chips), psychological states (e.g., beliefs, intentions, and goals), and motion capabilities (e.g., self-propulsion, agency, and contingent action).

These general definitions are also often followed in describing categorization in infants and young children. The terms "perceptual representation" and "perceptual categorization" are used to convey the idea that children attend to, and have knowledge of, the surface properties of objects and rely on these properties to classify things, and the terms "conceptual representation" and "conceptual categorization" imply that children have

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knowledge of, and presumably use, the nonobvious properties of objects, category relations (e.g., "this thing is an animal"), or what the *meaning* of something is (Mandler, 1992). According to a number of researchers (e.g., Quinn & Eimas, 1997; Mandler, 1992), a crucial difference between perceptually and conceptually based representations is that the latter allows inferences about the behavior and properties of objects or entities that are not available in the sensory array. This is not to say, however, that conceptual categorization excludes altogether any reference to perceptual cues. It seems difficult, if not impossible, to envisage a process whereby infants or children access conceptual knowledge about an object or entity without first identifying that object or entity on the basis of its perceptual appearance. This view is most vehemently argued by Mandler (1992, 1998, 2000, 2003) who posits two separate but connected processes in categorization. One of these processes involves perceptual categorization whereby an object or entity is identified based on its surface appearance, and the other involves conceptual categorization whereby the same object or entity is classified on the basis of its nonobservable properties (e.g., motion characteristics).

There are, however, a number of problems with the creation of a simple dichotomy of surface properties on the one hand and nonobservable properties on the other. Most importantly, to label perceptual properties as such based on their accessibility in the sensory array leaves questionable the identity of those only intermittently available to the perceptual system. Even though properties such as "acts as an agent" or "meows" are not continually present when one observes a cat, these properties become part of the representation for a cat through the visual and auditory systems. Thus it is possible, if not necessary, to ask at what point on the continuum between observable and nonobservable should a property be considered perceptual or conceptual? The resolution of this issue is crucial in determining the nature of the P2CS, but has thus far been generally overlooked in the developmental literature. My view, which is delineated in detail later in this chapter, is that a strong case can be made for considering all properties of objects perceptual in the sense that they enter the representational system through one or more of the sensory input systems and that the same basic associative learning mechanism underlies most, if not all aspects of this early knowledge acquisition. The consequence of this view is that there is no qualitative difference between representations for perceptual and so-called nonobservable information. For now, however, I think it important to bear in mind that in considering the P2CS one must not make staunch assumptions about the nature of perceptual and nonobvious information, their role in categorization at different ages, and the mental structures in which they are represented.

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Do Children Undergo a Stage-Like "Perceptual to Conceptual" Shift?

Although Piaget (1952) failed to provide a coherent explanation for how a sensorimotor infant develops into a conceptual infant, his influence on early cognitive developmental research led to an initial bias whereby any transition from a reliance on perceptual information to nonobvious information was viewed as a qualitative change in behavior (see also Bruner, Olver, & Greenfield, 1966). This assumption is without doubt understandable given the data that suggest that young infants are perceptually bound and older infants and preschool children are conceptually bound when making category membership decisions. For example, as highlighted at the beginning of this chapter there is evidence that children as young as 2½ years of age make inductions to perceptually dissimilar objects of the same category rather than to perceptually similar objects from a different category (Davidson & Gelman, 1990; Gelman & Coley, 1990; Gelman & Markman, 1987), and from around 4 years of age judge that nonobvious properties rather than the perceptual appearance of things determine what they are, or more specifically, how they should be labeled (Keil, 1989; see also Rips, 1989).

Despite this evidence of perhaps two (or more) transitions in behavior—one in infancy and one around 3 to 4 years of age—many researchers no longer believe that a "shift" from perceptual to conceptual categorization occurs; however, the reasons for this stance varies from researcher to researcher. Mandler (1992, 1998, 2003) argued that there is no need for a PC2S because although perceptual cues are important in the classification process, the "conceptual interpretation of perceptual arrays begins either at birth or as soon as stable perceptual schemas are formed in the first month or two" (Mandler, 2003, p. 106). According to Mandler, therefore, the primary basis for categorization is conceptual rather than perceptual virtually throughout the lifespan. A quite different view is taken by Oakes and Madole (1999, 2003) who claimed that perceptual and conceptual categorization essentially involve similar basic processes and are therefore intrinsically related. They argued that the observed change in infants' categorizing behavior emerges not because of the development of new categorization processes but instead is a result of ever-increasing access to different types of information via general development in cognitive, motor, and linguistic skills. For example, Oakes and Madole (2003) argued that the emerging ability to integrate spatiotemporal information allows infants to learn about functional properties at around 14 to 18 months of age. Similarly, Smith and Heise (1992) proposed that infants' experience with correlations in the environment causes them to pay greater attention to the attributes that are a part of those correlations. In this way, infants move from an unstructured "similarity space" to a

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distorted or structured similarity space that is weighted toward particular features and correlations of features.

Finally, it has been proposed that categorization is grounded in perceptual information not only during infancy but also throughout the lifespan. Elsewhere I have argued (Rakison, 2003; Rakison & Poulin-Dubois, 2001) as have Quinn and Eimas (e.g., Eimas, 1994; Quinn & Eimas, 1997, 2000) that concept acquisition is a process of continuous representational enrichment that relies on a perceptual system that is sufficiently sensitive and robust to allow infants to form categories that cohere because of similarity relations. Thus, young infants categorize on the basis of object features such as shape and texture (e.g., Jones, Smith, & Landau, 1991), functional parts (e.g., Rakison & Butterworth, 1998a; Rakison & Cohen, 1999), or facial features (e.g., Quinn & Eimas, 1996b), and later learn about many nonobservable properties of objects through language (Quinn & Eimas, 1997, 2000; Rakison, 2003; Rakison & Poulin-Dubois, 2001).

Quinn and Eimas and I differ, however, in our view of the very nature of perceptually and conceptually based representations. According to Quinn and Eimas (1997), perceptual categorization involves complex information but "does not include inferential information unavailable to the perceptual system" (p. 273). In contrast, conceptual representations are informative about object kinds and "presumably support numerous inferences about the nature and behavior of the thing represented" (p. 273). The implication of these statements is that although the general processes underlying the acquisition of representations remains constant throughout development, the very nature of perceptual and conceptual representations are fundamentally different. Although I agree with the general notion advanced in these statements, my stance is that infants' representations are "perceptual" throughout development in the sense that all information concerning things in the world is derived from sensory input. Moreover, I claim that these representations are capable of supporting some kind of inferences about the properties of objects or their behaviors at virtually all points in their evolution (Rakison, 2003; Rakison & Poulin-Dubois, 2001). For instance, there is evidence that infants as young as 3 months of age can develop *expectations* about the appearance of items on a screen (see Haith, Wentworth, & Canfield, 1993; Roberts, 1998). I present additional evidence from my laboratory later in the chapter to support this view.

As a more broad comment on the nature of the PC2S, it is worth pointing out that the movement away from a view of a stage-like shift in early category and concept development coincides with similar claims about developmental change in general. Siegler (1996) has forcefully argued that classic stage-based theories such as those of Piaget and Kohlberg, as well as more recent accounts such as those by Carey (1985) and Wellman (1990), sit uncomfortably with a good deal of developmental data. As an alterna-

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tive, Siegler argued that developmental change is better described and explained as continuous, with different strategies competing with each other as a series of overlapping waves. In this way, development is not a set of discrete "jumps" from one state to another but rather is best viewed as an increase and decrease in the frequency with which certain strategies, information, or processes are relied on.

In summary, although there is evidence that preschoolers and older children have knowledge about the nonobvious properties of objects, it is far from clear whether this knowledge results from a perceptual to conceptual shift of any kind. Certainly there is little support for the notion of a stage-like transition, and according to a number of theorists it is questionable whether a representational shift of any kind actually occurs. A number of important issues remain, however. Most notably it is unclear at what point in development, if at all, the change is made from perceptual to conceptual representations. Moreover, what causes this developmental change also remains moot. These issues are discussed, at least in part, in the following sections.

Do Infants and Children Rely on Perceptual Information or Nonobvious Information to Categorize?

Thus far in this chapter it has been taken for granted that infants tend to rely on surface properties to categorize—that is, have perceptual representations—and that older children tend to rely on nonobvious properties to categories—that is, have conceptual representations. I have already suggested that the distinction between perceptual and conceptual representations is more opaque than is generally assumed, mainly because of the argument that all information enters the representational system via one or more of the perceptual systems. At a different level of analysis, however, it is possible to examine the available evidence to assess whether it supports the claim that infants and young children categorize conceptually rather than perceptually.

Evidence of Conceptual Categorization and Induction in Infants. The notion that infants' representations involve conceptual information has been most strikingly argued by Mandler (1992, 1998, 2000, 2003; Mandler & McDonough, 1996, 1998b). According to Mandler, by around 12 months of age, infants no longer categorize objects because of their appearance but rather because of a shared *meaning*. For example, by around the end of the first year of life infants will group together various mammals because they are understood to be self-propelled entities that can act as agents. These conceptual categories are based on *image-schemas*, or *conceptual primitives*, that encapsulate crucial abstract characteristics of objects' spatial structure and

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movement (see also R. Gelman, 1990; R. Gelman, Durgin, & Kaufman, 1995). For instance, image-schemas that summarize animate entities represent self-motion, animate-motion (moving nonlinearly), causing action at a distance, and agency. Mandler (1992, 2000) proposed that these image-schemas are constructed through an innate process of *perceptual analysis* that recodes aspects of the perceptual display into a more abstract, accessible form.

The evidence that Mandler used to support this theory comes from three separate experimental procedures used with infants. Data from sequential-touching and object-examining tasks revealed that infants in the second year of life formed categories when stimuli possessed low within-class perceptual similarity—for example, animals, vehicles, or furniture (Mandler & Bauer, 1988; Mandler et al., 1991)—and infants as young as 9 months of age categorized as different items that were perceptually similar such as planes and birds (Mandler & McDonough, 1993). In the inductive generalization task infants observe an action and are allowed to repeat that action with two objects, one drawn from the same category and one drawn from a different category than the model exemplar (Mandler & McDonough, 1996, 1998b). As outlined earlier, using this methodology it has been shown that infants as young as 9 to 11 months generalize behaviors like drinking and sleeping to animals and behaviors such as "starting with a key" and "gives rides" to vehicles. Moreover, when infants were presented with two objects from the same superordinate category to repeat an action (e.g., a dog and a cat after seeing a dog "drink" from a cup), they were just as likely to use either exemplar, a finding that was interpreted to mean that infants understand that both objects were capable of that action.

There are, nevertheless, both empirical and theoretical grounds to question Mandler's view that infants categorize conceptually within the first year of life. At a theoretical level, Mandler's (1992, 1998, 2000) theory has been questioned for its failure to specify how the process of perceptual analysis—whereby information available in the perceptual array is translated into a simpler, conceptual image-schema—might operate. This process, which is the crux of Mandler's theory, is as yet laid out only in general terms that fail to provide sufficient information to allow for a direct empirical test to be made. An additional problem with Mandler's formulation is that as a consequence of the notion that there are distinct mechanisms that deal with perceptual and conceptual categorization, the appearance of objects is represented separately from their meaning. That is, "image schemas representing biological motion, self-motion, and contingent motion would be independent of the actual appearance of the movement of any particular object" (Mandler, 2000, p. 20). One cannot help but wonder whether it is necessary to posit such a dual-process framework when a single, integrated representational mechanism offers a more parsimonious and just as viable

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framework for concept development in infants and adults (Goldstone & Barsalou, 1998; Quinn, Johnson, Mareschal, Rakison, & Younger, 2000). Finally, it is questionable whether the process of perceptual analysis differs from perceptual categorization of movement patterns (Quinn & Eimas, 2000). There is no compelling reason why movement patterns of objects need to be abstracted in the way formulated by Mandler (1992, 2000): It is just as reasonable to suppose that these perceptually available patterns of movement are associated with individual exemplars, and then later are generalized to other category members.

At an empirical level, it is unclear why it is necessary to posit that infants have early conceptual knowledge about objects' meaning in order to explain the sequential touching, object examining, and generalized imitation data that Mandler presents (e.g., Mandler & Bauer, 1988; Mandler et al., 1991; Mandler & McDonough, 1993, 1996, 1998a, 1998b). In many cases there is evidence that perceptual similarity can play the role ascribed to conceptual knowledge in such experiments. For example, evidence from sequential touching studies by Rakison and Butterworth (1998a) revealed that 14- to 18-month-old infants' ability to form superordinate categories such as animals, vehicles, and furniture is guided by the presence or absence of object parts such as legs and wheels. For example, infants as old as 14 and 18 months categorize animals that possess legs as different from vehicles with wheels but they are unable to do so when animals possess no legs and vehicles possess no wheels or when animals and vehicles possess both legs and wheels.

There are also data to suggest that infants' inductive generalizations in the task used by Mandler and McDonough (1996, 1998b) may be guided more by perceptual similarity than knowledge about the specific actions of animates and inanimates. Using the same general procedure as that in the studies by Mandler and McDonough (1996, 1998b), I have found that infants will repeat the observed actions with a category exemplar from the "wrong" category if the experimenter models the action with an exemplar from that "wrong" category (Rakison, 2003). For example, if an experimenter models a cat going up a set of stairs, 14- and 18-month-olds will repeat the action with a novel animal; but if the experimenter models a car going up a set of stairs infants in the same age groups will repeat the action with a novel vehicle (Rakison, 2003).

To investigate further infants' knowledge of motion properties and their behavior in the task, I recently developed a novel version of the inductive generalization procedure and tested infants with two simple motions of animates (a dog walking/hopping and a bird flying) and two simple motions of inanimates (a car rolling and a plane flying) (Rakison, 2004). In the modified version of the task, infants were presented with four objects rather than two in the baseline and generalization phases. One of the stimuli was

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drawn from the same superordinate domain as the model and shared overall similarity and functional parts with that exemplar (SPSC: same parts, same category). For example, when the cat was the model exemplar the SPSC exemplar was a dog. Another test stimulus was drawn from the same superordinate domain as the model but did not share overall similarity or functional parts (DPSC: different parts, same category). For example, when the cat was the model exemplar the DPSC exemplar was a dolphin. A third stimulus was drawn from a different superordinate domain than the model but had the same functional parts as the model (SPDC: same parts, different category). For example, when the cat was the model exemplar the SPDC exemplar was a table (with four legs). Finally, one stimulus was drawn from a different superordinate domain and shared no obvious functional parts with the model (DPDC: different category different parts); when the cat was the model the DPDC exemplar was a car. A full list of the stimuli used in the task is presented in Table 6.1, and the stimuli from the cat "rolling" condition can be seen in Fig. 6.1.

This experimental design made it possible to explore the basis for infants' choice of objects to generalize the observed actions as well as whether knowledge for motion properties develops in parallel for the domains of land- and air-movement. Moreover, in separate experiments the motions were modeled to infants with either the appropriate exemplar (as in Table 6.1) or a simple U-shaped block (see Fig. 6.2). The idea behind the use of a U-shaped block as a model was to give infants as little information about the identity or category of the objects that performed the motions in question and thereby test more stringently their knowledge of those motions.

To simplify matters, I discuss infants' behavior only in conditions in which land-based motions were modeled (i.e., hopping and rolling). In the experiment in which the events were modeled with the appropriate exemplar, it was found that infants at 14 months repeated the observed action with the SPSC exemplar only marginally more than any other exemplar, suggesting that they would choose only those stimuli that were perceptually

TABLE 6.1
Stimuli Used in Inductive Generalization Studies in Which Events Were Modeled With an Appropriate Exemplar (as Shown Here) or an Ambiguous U-Shaped Block (Rakison, 2004)

		Test Stimuli			
Motion Events	*Model*	*SPSC*	*SPDC*	*DPSC*	*DPDC*
Vehicle rolling	Car	RV	Stroller	Boat	Cow
Animal walking	Cat	Dog	Table	Dolphin	Truck
Vehicle flying	Plane	Space shuttle	Dragonfly	Car	Duck
Animal flying	Eagle	Parrot	Plane	Dog	Grasshopper

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FIG. 6.1. Example of stimuli used in the cat "rolling" condition (Rakison, 2004).

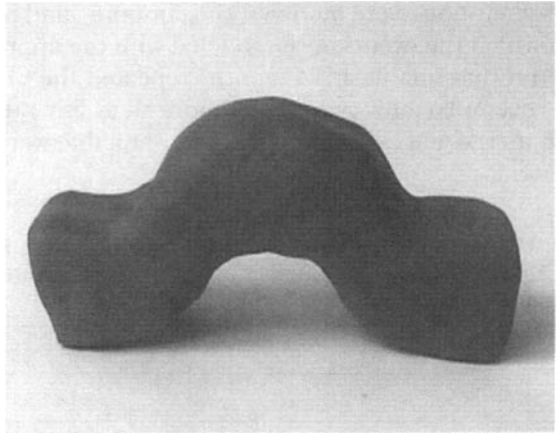


FIG. 6.2. The ambiguous block stimulus used to model events.

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(or conceptually) similar to the model. In contrast, infants at 18 months chose the SPSC exemplar and SPDC exemplars to demonstrate the actions. Infants at 22 months, however, tended to repeat the observed motions only with the perceptually and conceptually similar exemplar (SPSC). These findings imply a developmental trend whereby infants at 14 months have little or no basis for generalization, infants at 18 months relate object parts with specific motions, and infants at 22 months have generalized motion properties to whole objects but only those that are similar overall to the model (e.g., from a cat to a dog but not a dolphin).

The experiment in which the actions were modeled by the experimenter with the block exemplar were even more revealing about 18- and 22-month-old infants' developing representations for animates and inanimates (Rakison, 2004). In the experiment in which the land tasks were modeled with a U-shaped block, 18-month-olds chose the SPSC and SPDC exemplars to repeat the observed action, whereas 22-month-olds chose the SPSC and DPSC exemplars to hop or roll. This pattern of results suggests that infants brought some knowledge of the motion characteristics of animates and inanimates with them to the task: Infants were able to demonstrate actions even when no information was presented about the items that engage in those actions. That the 22-month-olds behaved differently in this experiment in comparison to those of the same age who saw an appropriate exemplar model the same action suggests that infants are affected somewhat by the stimulus used in the experimental situation (see also Rakison, 2003). That is, infants may have been directed to imitate only actions with the exemplars that were similar to the model, and when no information was given about the model, infants were more unconstrained in their choice.

In my view these studies have a number of implications for the discussion at hand. First, they support the notion that infants at least as old as 18 months do not generalize the actions they observe in the inductive generalization procedure on the basis of nonobservable properties or category membership. Instead, the data presented here, which are consistent with my earlier work (see Rakison, 2003; Rakison & Poulin-Dubois, 2002), suggest that 14- and 18-month-old infants' knowledge of the motion properties of objects may be based primarily in an association between an action or motion and the causally relevant functional parts involved in that action or motion. In contrast, older infants, those around 22 months, appear to have extended this association to whole objects or categories of objects (see also Jones & Smith, 1993; Quinn & Eimas, 1997). This conclusion is supported by infants' behavior across the two experiments described here, but the most compelling evidence was found in the block-model study. In that task infants at 18 months in the land-task chose the objects that possessed the appropriate parts for the motion (e.g., they made the cat and table "walk"), whereas infants at 22 months chose the two objects that belong to the ap-

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propriate category for the motion (e.g., they made the cat and dolphin "walk"). Ongoing research in my laboratory is exploring how infants start to generalize motion cues from causally relevant parts to other object features or whole objects.

Evidence of Conceptual Categorization and Induction in Preschoolers. The notion that older children base their categorization and inductions on knowledge about nonperceptual properties has also come under scrutiny in recent years. In particular, Smith and her colleagues (e.g., Jones & Smith, 1993; Smith & Samuelson, 1997; Thelen & Smith, 1994) argued that in studies like those conducted by Gelman, in which perceptual properties are pitted against category membership, perceptual aspects of the stimuli are impoverished compared to their real world appearance. This, in addition to the fact that the objects are typically labeled with familiar names—snake, worm, cat, skunk, for example—means that children have to make a category induction either on the basis of nonobvious properties and a label or on the basis of a line drawing. Similarly, in the transformation studies by Keil (1989), children are presented with a line drawing and a considerable amount of information about the entity's nonobvious properties. Perhaps children interpret this bias in the amount of information given as a suggestion that, in fact, the object is defined by its nonperceptual properties, and therefore they tend to choose nonobvious properties to decide what kind of thing something is. Indeed, there is evidence that children 8 years and younger will generalize a trait to a perceptually identical drawing rather than one with the same label (Farrar, Raney, & Boyer, 1992). Finally, Jones and Smith (1993) suggested that children and adults in the induction and transformation studies may use general knowledge about biology, the validity of stories, and even the psychology of experiments, rather than knowledge about skunks, raccoons, or worms per se. In other words, judgments about category membership in such tasks may be "assembled from multiple kinds of knowledge that include perceptual, nonperceptual, linguistic and even social referents" (p. 137).

The crux of Jones and Smith's (1993) claim is that perception and language play roles equal to nonperceptual information in naming, categorization, and conceptual knowledge. It should be borne in mind, however, that although the arguments by Smith and colleagues are forceful, there is little in the way of data to support or refute them. Jones and Smith (1993; see also Jones et al., 1991) presented evidence that features such as overall shape, eyes, and texture are important cues in categorization and label extension; however, whether there is some stable representational core that predominates in categorization, induction, and language, and what information is stored at that core, remains unknown. And indeed, an inherent problem with studies in which perceptual and conceptual information are

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pitted against each other is that it is difficult, if not impossible, to balance the presentation of such cues. As a result, for any given finding that falls on one side of the argument or the other—that is, data that suggest that young children use perceptual information or nonobvious information to categorize—it is possible to make the case that the different types of information were not equally salient in the task. Nonetheless, it is important to consider seriously the empirical data and theoretical arguments outlined here because they at least bring into question whether infants and preschoolers rely on nonobvious properties in categorization, induction, and language.

Is Perceptual Information Less Reliable Than Nonobvious Properties as a Basis for Categorization?

The discussions in the previous sections highlight three points: First, it is difficult to discern or argue for a qualitative difference at a representational level between perceptual and nonobvious properties; second, there is little evidence of a qualitative "shift" from one kind of representation to the other; and third, it should not be tacitly assumed that infants and children rely on nonobvious properties to classify, or in other words, that they engage in what has been called conceptual categorization (Mandler, 1992). But what of the claims that conceptual representations must underlie categorization because nonobvious properties are more reliable and less misleading about category membership than perceptual, surface properties? The theoretical argument for this claim rests on the representational structure of information. It is putatively held that the core of concepts are stable in that they are built from a list of propositions that are unchanging, static representations (Medin & Ortony, 1989); for instance, any number of people would know that dogs give birth to puppies rather than kittens. A corollary is that perceptual similarity is inherently unstable in that the aspect of similarity to which adults attend varies depending on the context (e.g., Barsalou, 1989; Jones & Smith, 1993); thus in one context kittens might be considered more like lions, and in another they might be considered more like puppies. If both of these premises are accepted, then, according to some, it logically follows that perceptual information cannot be at the core of stable representations and therefore nonobvious properties must play such a role (Medin & Ortony, 1989).

Within this framework, it has been claimed that there is a nonarbitrary, causal relationship between surface and deep properties whereby the central nonobvious attributes of a thing constrain and even generate its surface attributes. According to Medin and Ortony (1989; see also Brown, 1989; Rips, 1989; E. E. Smith, Medin, & Rips, 1984), this causes children and adults to be drawn to surface similarities that in turn lead them to more central conceptual properties. For example, a child might find animal legs

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perceptually salient and consequently be guided to internal properties such as animacy, support, and so on. Thus, despite perceptual similarities, objects are judged to be members of a larger category by virtue of qualities that are invariably not perceptual, or they are defined by some underlying nature that makes them what they are. The result is what Medin and Ortony (1989) called *psychological essentialism.* They defined it as the belief that objects of a category share some sort of common essence or are "the same kind of thing," and claimed that it is at the core of concepts and therefore category membership decisions. In support of such a view one could refer to the work of Gelman, Keil, and others that was discussed earlier in the chapter.

This line of reasoning is certainly persuasive and has been highly influential within the field. Nevertheless, there are a number of reasons why this theoretical view may not accurately describe the nature of developing representations. First, that adults will attend to different aspects of surface similarity depending on contextual cues should not be considered as an inherent weakness—as appears to be Medin and Ortony's (1989) view—but instead can be viewed as a strength in terms of the dynamic and flexible way in which perceptual information (as well as other kinds of information) can be used differently in a variety of situations (see also Jones & Smith, 1993; L. B. Smith, 2000; Thelen & Smith, 1994). Second, it remains to be seen whether the claim that the core of representations are a stable list of propositions is a veridical one. Although this view is probably held implicitly by many of those within Cognitive Science who study concepts and concept acquisition, there is no direct evidence that it is an accurate portrayal of representational content. Indeed, in combining both of these last two points, Jones and Smith (1993) claimed that: "What we call 'categories' and 'concepts' are the emergent products of multiple knowledge sources in specific task contexts. By this view, there is no set intension (definition in the head) or extension (category in the world). Both are transient and emergent in the task at hand" (p. 136).

Finally, and somewhat more problematic for Medin and Ortony's view, it is important to reflect on the advantage of attention to nonobvious properties, or in other words, to think about categorization in terms of a simple cost-benefit analysis. I do not doubt that surface properties are constrained and caused by nonobvious properties or that such nonobvious properties are in some sense a more reliable cue of category membership. Thus it is beyond argument that tigers' stripes have a genetic foundation and that tiger DNA is a better predictor than stripes that an animal is a tiger rather than, say, a zebra. However, my claim is that although as adults we may know the causal relation between deeper, nonobservable properties and surface properties, there is no reason to access this knowledge when surface properties will suffice in category membership judgments.

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Instead, I propose that perceptual cues provide sufficient information for accurate categorization in all but a few cases, and the cost of accessing information about conceptual properties generally outweighs the benefits. To return to the example I just used, if a tiger-looking big cat were to walk into my office would it be more parsimonious to decide from its appearance—large, striped, with big teeth—that this animal has the potential to attack and kill me or would it be necessary for me to refer my knowledge of nonobvious properties such as whether it has tiger DNA or its parents were tigers? I think that few would argue that accessing nonobvious properties would be helpful in such a situation. Let us consider a more everyday example. My Honda Prelude looks very much like a car. It has four wheels, windows, seats, furry dice, and all the other amenities that a car should have. Would you need to determine that it has an engine, requires gas, and can actually function as a car in order to call it a car? Or would you assume that if it looks like a car, it is, most likely, a car?

So why is it that in studies such as those by Gelman and Keil, children are more likely to use nonobvious properties than surface properties to make category membership judgments? I propose that what emerges around the preschool years is not a perceptual to conceptual shift at all but rather an understanding that nonobvious properties—those that cannot be seen—are more highly correlated with category membership than surface features. This does not mean that surface features are poor predictors of what kind of thing an object is, but instead that nonobvious properties are more strongly, if not perfectly, related to category membership. This newfound knowledge does not mean, however, that the default basis for categorization becomes nonobvious properties; surface properties, for the reasons given in the previous paragraph, continue to play a primary role in everyday category decisions. But, when perceptual cues are directly pitted against nonobvious properties—as they are in the studies by Keil, Gelman, and others—children will chose the latter over the former because of this newly developed assumption about their ontological predictive power. This interpretation of preschooler's behavior is consistent with Siegler's (1996) view of conceptual change as overlapping ways. Thus, it is not that one representational content or structure replaces another but rather that a new strategy emerges for categorizing objects, events, and properties, and this strategy competes with those already available to the young child and is deemed most productive under certain conditions. To test this hypothesis, I have recently investigated the role of similarity and nonobvious properties (e.g., has a round heart, likes people) on inductive inference by preschoolers and adults (Hahn & Rakison, 2003). Hahn and I examined category membership decisions in the context of varied statistical information about the distribution of surface and nonobvious properties. Participants were taught new categories of animals, birds, and insects that varied in the degree to

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which nonobvious or perceptual properties predicted category membership. The results showed that preschoolers as well as adults use distributional information to inform their inductive inferences; that is, both groups used surface properties or nonobvious properties when each was the better predictor of category membership, and they did so whether or not labels were present.

Some critics of this view may argue that what is at issue here is the difference between judging something to be a member of a category and recognizing it as an exemplar of a category. In other words, it could be claimed that perceptual cues are used to determine what something is whereas nonobvious cues are used to determine what the meaning of something is (e.g., Mandler, 1992, 2000, 2003). In many cases, however, these two aspects of object processing are difficult, if not impossible, to discriminate, and it is unclear whether it is necessary to access knowledge of nonobvious properties in everyday category membership decisions.

What Mechanisms Underlie the Acquisition of Knowledge About Perceptual Properties and Nonobvious Properties?

A final issue that is integral in any discussion of the perceptual to conceptual shift, and in any discussion of cognitive development in general, concerns the mechanisms that underlie the acquisition of representations early in life. I use the term *mechanism* here to refer to the mental processes that allow or cause developmental change (see Siegler, 1996). Somewhat surprisingly, a focus on the mechanism underlying change is commonly absent from discussion of conceptual developmental. For example, those who support the theory theory view often overlook the issue of developmental change because it is believed that relatively sophisticated concepts are present early in life (Gelman & Koenig, 2003).

In recent years, however, a focus on mechanism has become more prominent in the work of those who study infants' representations. Leslie (1984, 1988, 1995), for example, presented a nativist and *modular* view of infants' developing representations of objects. He proposed that infants are born with a three-part theory of Agency, with separate brain mechanisms for each, which process the mechanical, intentional, and cognitive properties of agents. Thus, distinct mental modules direct infants' attention toward different kinds of events and allow them to interpret those events; for example, a *theory of body*, or *ToBy*, biases infants to interpret agents as having an internal and renewable source of energy, and a *theory of mind mechanism* (ToMM) interprets an agents' action as goal directed or involving intentionality. Elsewhere, I have argued that there are a number of problems with such a modular view (Rakison & Poulin-Dubois, 2001). First, it is not clear why these modules are not initially functional but instead must be

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"triggered" at some, generally unspecified, point in the first years of life. For example, infants do not perceive actions as causal or goal-directed until around 6 to 7 months of age (Oakes & Cohen, 1990; Leslie & Keeble, 1987; Woodward, 1999). Second, it remains an open question how distinct modules deal with information that involves both mechanical and intentional action such as when a hand reaches for a toy. Mandler (1992, 1998, 2000, 2003) offered perhaps the most influential proposal of the representational change in infancy. As was outlined earlier, Mandler posited that from very early in life infants possess two representational systems, one that supports perceptual categorization and one that supports conceptual categorization. The perceptual categorization system is involved more in object recognition whereas the conceptual categorization system allows infants to classify objects based on their meaning.

A third view concerning the mechanism for representational change, which was also outlined earlier, is that the information incorporated into infants' and children's concepts is inherently perceptual in nature and that representational development occurs through gradual, incremental learning (Eimas, 1994; Quinn & Eimas, 1997, 2000; Rakison & Poulin-Dubois, 2001; Rakison, 2003). The claim is that concept acquisition is a process of continuous representational enrichment that relies on a perceptual system that is sufficiently sensitive and robust to allow infants to form categories that cohere because of similarity relations. The development of language allows the acquisition of knowledge about the nonobservable characteristics of objects—in particular, biological functions like reproduction and other internal properties—through formal and informal tuition. Moreover, it has been postulated that children are able to acquire information about objects through language after very few exposures, and this information may be generalized broadly across category members (Rakison, 2003). This is a form of what has been labeled *fast mapping* (see Bloom, 2000) although in the case outlined here it is not that infants connect a label to a referent but rather they connect an object property to a category. For instance, children may learn that dogs have hearts after hearing this proposition only once or twice, and they would assume that many, if not all dogs possess hearts. In combination, these processes lead, ultimately, to a representation for animals that include shape, texture, parts, facial features, and less obvious, nonobservable properties.

As discussed earlier, my view differs from that of Quinn and Eimas in a number of ways. Most notably, Quinn and Eimas argued that only conceptual representations, and not perceptual ones, can support inferential processes whereby infants can use information unavailable to the perceptual system. In contrast, I do not support the distinction, either at a semantic or theoretical level, between perceptual and conceptual representations, and I believe that even the earliest representations can support inferential proc-

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esses in the form of *expectations* that result from the association between static and dynamic cues. Thus, the perception of one component of the relationship activates an expectation about the existence of the other component (Rakison, 2003; see also Haith et al., 1993; Roberts, 1998). In the case of properties that are available only intermittently in the perceptual array such as motion properties, the result is a representation of an associative link that embodies what has come to be labeled "nonobvious" knowledge; however, there is no need to apply the kind of rich description to these representations that has been suggested in previous formulations of concept development (e.g., Leslie, 1995; Mandler, 1992). For the kind of expectations outlined here, what is associated is a perceptual feature—legs, for instance—with an intermittently available motion characteristic such as self-propulsion.

An additional divergence from Quinn and Eimas' (1997, 2000) theory is that I have proposed the early presence of a number of *attention biases*—though other terms such as *constraints* or *learning enablers* could just as easily describe them—that direct infants to causally relevant information available in the perceptual array. The presence of these attention biases means that not all information available in the perceptual array is equally salient to infants, and consequently the theory is not undermined by the argument of Original Sim (Keil, 1981). These biases include, but are not limited to, attention to motion, both at a global and a local level, and attention to the static and dynamic features of entities and objects. An attentional system that is drawn to motion is most likely an evolutionary adaptation that helps humans, as well as other species, to detect the presence of potential prey or a potential predatory threat. Indeed, there is evidence that newborns prefer moving objects over static objects (Slater, 1989), that 4-month-olds' perception of object unity for a partly occluded object depends on common motion (Kellman & Spelke, 1983), and that by 3 months of age infants prefer a human moving point-light display to an unstructured point-light display (Bertenthal, 1993). Moreover, motion cues continue to play an important role throughout the lifespan. It has been found that adults and infants detect an object's properties more easily when it moves than when it is stationary (Burnham & Day, 1979), and that 4- and 7-year-olds as well as adults base their category judgments for animals on motion characteristics rather than shape when those two attributes are pitted against each other (Mak & Vera, 1999).

Support for this view of representational development in infancy comes from a number of studies with a variation of the Switch design (Gentner, 1978; Younger & Cohen, 1986) within the habituation procedure. In the basic Switch design, infants are habituated to two events, one in which feature A^1 is paired with feature B^1 and one in which feature A^2 is paired with feature B^2. Once habituated to these stimuli, infants are presented with a trial in

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which the feature pairings are switched—for example, feature A^1 is shown with feature B^2—and a trial in which the feature pairings remain unchanged from that shown during habituation. This design is especially ideal for examining infants' ability to learn correlations because relatively long looking time to the "switch" test trial compared to a familiar trial can only be caused by infants' detection of a novel feature pairing and not because of the introduction of a novel feature. Using this basic design, it has been shown that 7-month-olds can detect correlations among the static features of schematic animals (Younger & Cohen, 1986), 14-month-olds can associate a label with a moving object (Werker, Cohen, Lloyd, Casasola, & Stager, 1998), and 14- to 18-month-olds can associate an object's form with its function (Madole, Oakes, & Cohen, 1993; Madole & Cohen, 1995). In the novel variation of this design described here, infants were habituated to events in which there were three variables rather than two (Rakison & Poulin-Dubois, 2002). Thus, infants were habituated to pairings $P^1B^1M^1$ and $P^2B^2M^2$ (P = parts, B = body, M = motion path) and then presented with test events that violated these correlations. The strength of such a design is that it allows investigation of infants' ability to detect and encode correlations as well as their predilection to attend to certain correlations and ignore others.

In one series of studies, 10-, 14-, and 18-month-old infants' were habituated to two events in which an object moved across a screen. Each object consisted of a unique body (red oval-shape or blue pot-shape), a pair of unique moving parts (yellow cigar shapes that moved horizontally or green diamond shapes that moved vertically), and a unique motion path (rectilinear or curvilinear). Infants might see during habituation, for example, a blue-bodied object with yellow horizontally moving parts travel along a linear trajectory and a red-bodied object with green vertically moving parts travel along a curvilinear trajectory. During the test phase of the experiment, infants were shown one trial in which the parts were switched in relation to the pairing seen during habituation (e.g., $P^2B^1M^1$), one trial in which the body was switched in relation to the pairing seen during habituation (e.g., $P^1B^2M^1$), one trial in which the motion was switched compared to that seen earlier (e.g., $P^1B^1M^2$), and one trial that was the same as that seen during habituation. For example, given the two habituation stimuli just described, in the parts switch infants might see the blue-bodied object travel linearly but with green horizontally moving parts, in the object switch they might see the blue-bodied object with green vertically moving parts travel along a curvilinear trajectory, and in the motion switch they might see the red-bodied object with green vertically moving parts travel along a linear trajectory.

The results of the experiments were not only consistent with previous research in this area (e.g., Madole & Cohen, 1995; Werker et al., 1998), but they also supported the idea that correlations in which one or both cues are

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dynamic are privileged in infants' processing of events. Infants at 10 months of age failed to learn any of the correlations, and a subsequent experiment revealed that they could process independently the static bodies of the objects but not the dynamic parts or motion trajectory of the object. In contrast, infants at 14 months attended to the correlation between an object's parts and its motion trajectory, as evidenced by a significant increase in looking time to the part switch in comparison to the familiar test trial. In addition, 18-month-old infants recovered visual attention to all three switch test trials suggesting that they had encoded all of the correlations among the features in the event; that is, they had learned that objects with particular parts and a particular body moved in a certain way.

To test whether the dynamic nature of features is crucial in capturing infants' attention to causally relevant relations involving those features, in a separate experiment 14- and 18-month-olds were presented with the same habituation and tests events as those previously described, but in these events the features of the objects did not move. The results of the experiment revealed that 14-month-old infants under these conditions no longer learned the correlation between object parts and motion trajectory (or any other correlations in the events) and 18-month-olds attended only to the correlation between parts and motion trajectory. The behavior of the 18-month-old group in this experiment suggested that they expected that the external parts of an object are related to the motion characteristics of that object. Finally, in more recent work in my laboratory (Rakison, in press), I used a similar design to examine the age at which infants can use the relationship between dynamic parts and a motion trajectory to form categories. This work has revealed that it is not until 22 months that infants form categories based on these correlations, and they will not form a category based on other, less salient correlations such as that between a motion path and a body type or a body type and part type.

Taken together, these studies suggest that the movement of causally relevant parts may help infants to distinguish such features from other, less significant features. Moreover, the data suggest that an associative learning mechanism underlying the ability to encode correlations among dynamic features may come online at some point early in the second year and is not limited to the linguistic domain (Werker et al., 1998). Perhaps more impressively, these data suggest that infants are selective in the correlations to which they will attend. In all likelihood, this selectivity acts as a learning constraint by directing infants to discover, and learn about, those relations that exist between features that are causally relevant. Infants would learn first, for example, that things with legs move one way and things with wheels move another way, and then later they would generalize this correlation to other object features possessed by the objects in question (e.g., faces, tails, seats, headlights, and so on).

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SUMMARY AND CONCLUSIONS

In this chapter I examined five separate but interconnected themes concerning the perceptual to conceptual shift in early childhood. Based on a review of the literature of related theoretical and empirical work on these themes, a number of conclusions have been made about the nature of early representational development. I outline these conclusions as follows.

First and foremost, I argued that there is no logical or empirical reason to differentiate between perceptual information and that which is nonobservable because both are acquired through the same domain general associative learning mechanism (Rakison, 2003; Rakison & Hahn, in press). Thus, in my view the same processes are involved in learning that a dog has legs, that a dog is self-propelled, that a dog has a heart, and that a dog is called a "dog." In each case, the information is conveyed through a perceptual input system—in the examples given here, the visual and linguistic systems—and associative processes are sufficient to explain how each becomes part of the child's mental structures. Moreover, I have made the claim, and presented evidence to support this claim, that infants' earliest representations can support a simple sort of inference that is best thought of as an expectation. Once an infant learns that things with legs are also self-propelled, they would expect an object with legs to move on its own (even if it were not moving) and they would expect that things that move on their own have legs (even if those legs cannot be seen).

A direct implication of these claims is that no perceptual to conceptual shift occurs in infancy or early childhood. That is, it is redundant to think about a transition whereby infants and younger children possess perceptual representations—which do not support inference or carry any meaning about an object—and older children and adult possess conceptual representations—which do support inference and support meaning. Instead I argued that infants' earliest representations are fundamentally similar, though impoverished versions of the representations found in older children. Moreover, I have posited that the putative qualitative shift found in categorization studies such as those by Keil, Gelman, and others is observed for two reasons. First, it is the case that surface and nonobservable properties are presented in these studies in such a way that the latter is more salient with regard to the amount of information given and the quality of that information (Jones & Smith, 1993). Second, it may well be that nonobvious attributes, which can be considered perceptual in the sense that they are learned via one of the sensory modalities, are primary in the kinds of studies already mentioned because they are perfectly correlated with category membership whereas other, more surface attributes are less so. For instance, all birds have bird DNA but not all birds can fly. This is not to say, however, that the categories that children and adults form in the real world

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will cohere because of such "nonobvious" properties; after all, if something has wings and feathers it is a pretty good bet that it is a bird. But in studies where two attributes are pitted against each other—irrespective of the methodological issues discussed earlier—it is perhaps not surprising that children might favor the one that is more invariantly associated with category membership. What might emerge in the preschool years is not, therefore, a predilection to rely on nonobvious cues, but rather the assumption that nonobvious cues are more strongly correlated with category membership than are surface properties.

In summary, in this chapter I presented theory and evidence to support the notion that it is erroneous to think of representational development as undergoing some kind of transitional shift either in infancy or the preschool years. Rather, perceptual information provided by the different sensory input systems, including language, is continually added to the earliest representations through associative learning, and as sensitivity to different sorts of information emerges, so this gradual process gives the appearance of qualitative, representational change. The available evidence goes some way to support this position, though clearly further research is needed to substantiate the claims made here, including that from other disciplines within Cognitive Science such as computational modeling (see Rogers & McClelland, chap. 14, this volume). Nonetheless, the theory posited here is very much in line with current thinking in early cognition that stresses an infant who is reliant on information available in the perceptual array and uses that information during category and concept formation (e.g., Eimas, 1994; Jones & Smith, 1993; Mareschal & French, 2000; Quinn & Eimas, 1996a; Younger & Fearing, 1999, 2000). The next step, and one that I think is destined to prove fruitful, is to determine more explicitly how and when these early representations evolve into the more detailed and complex ones that we, as adults, possess.

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